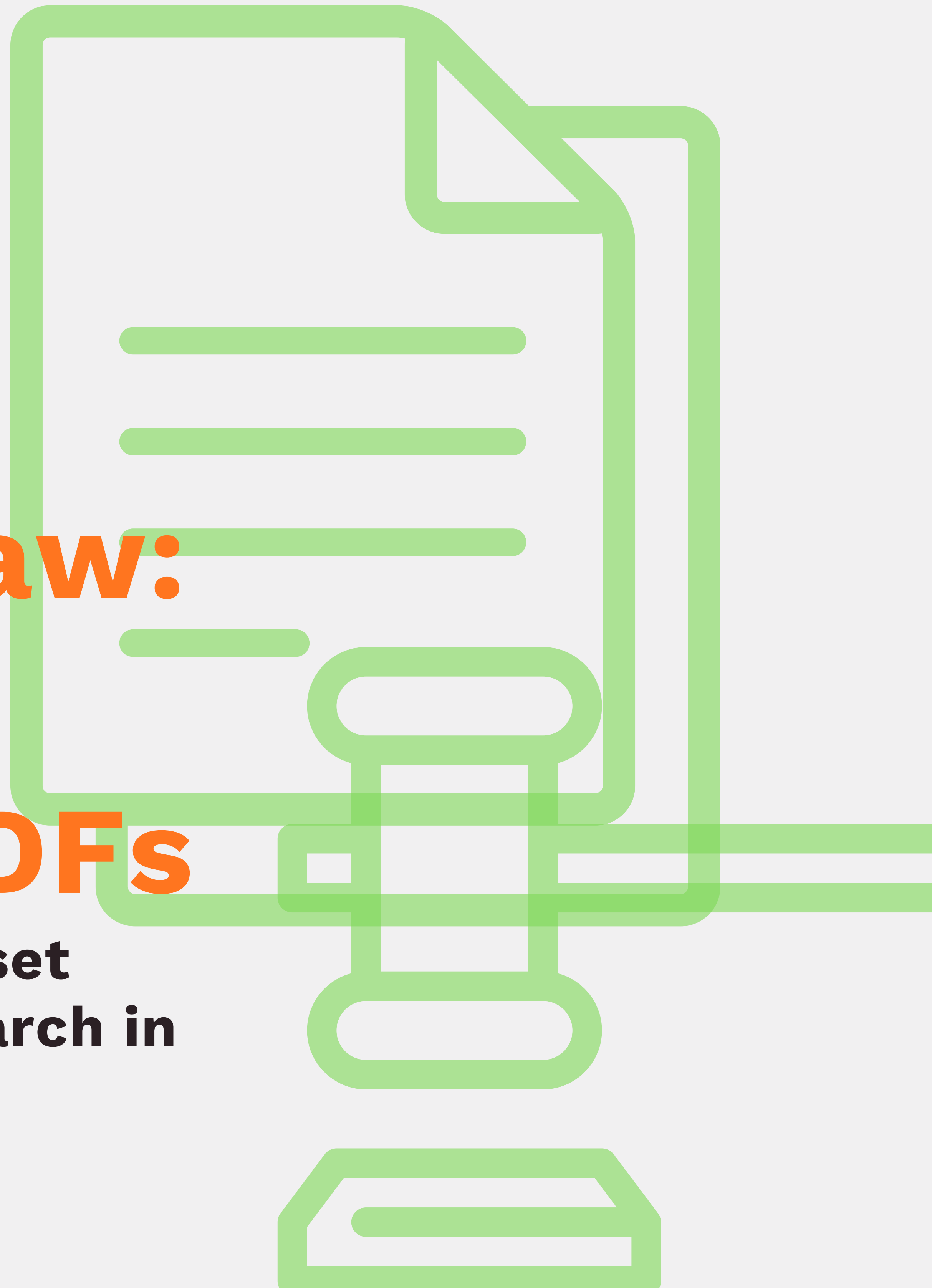
Written in English for clarity — some technical terms don’t translate naturally yet

Unlocking Indonesian Law: ~250,000 Regulation PDFs



Introducing a Massive, Dataset Collection for Legal AI Research in Bahasa Indonesia.

Afza Azzindani

The Challenge of Legal NLP in Indonesia

- **The Problem:** Training state-of-the-art AI for Indonesian legal systems requires a vast, clean, and organized corpus of primary source documents.

- **The Solution**: I built a high-volume, cloud-based pipeline to aggregate **~250,000** Indonesian Regulation PDFs from over **350,000** public URLs.

- **The Result**: The largest collection of its kind, enabling foundational research in regulation-aware AI and document classification in Bahasa Indonesia.

- **Disclaimer**: All documents were collected from publicly accessible government sources for research purposes only. Users must verify the accuracy and legal validity of the content before applying it in practice.

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Time & Cost-Saving Data Collection Responsibly

Web scraping works by mimicking human browsing, but with extreme efficiency, making large-scale data collection feasible:

Process	Manual Browsing	Automated Scraping
Action per Doc	Navigate, click link, wait for load, save file	Send HTTP request, parse PDF link, download immediately
Time per Doc (Est. in second)	**~30**	**~12**
Workload	Huge physical & mental fatigue, can't scale	Very low manual effort — can be done while sleeping
Cost (Est. per hour)	**2 USD**	**0.05 USD**

- **Ethical Review**: I adhered to R&D best practices by only targeting publicly accessible legal documents and implemented rate limiting to prevent server overload.

Strategic Cost & Compute

I chose a strategic **cloud approach** because it drastically lowered my cost for this high-volume R&D project

Constraint	Local Machine (PC, Laptop)	Cloud Strategy (Google Colab & Hugging Face)
Computing cost	Requires purchase of **high-end hardware** for parallel scraping	**Low Cost**: Leveraged Google Colab's free instances (with limited, but sufficient, computing power)
Storage cost	Requires hundreds of GB of dedicated **local storage**	**Zero Cost**: Used Hugging Face's free public dataset storage
I/O bottleneck	**Slow and unstable Indonesian internet** (~45 Mbps) causes frequent connection drops and ruined downloads	Access to **high-speed cloud bandwidth** ensures consistent download rates and stability
Uptime/ maintenance	Prone to **PC crashes**, heat issues, and power interruptions over 200+ hours	**High uptime**, but required the manual workaround (e.g., keeping the screen active) to prevent free instance disconnection

- **Final Architecture**: The workload was split across **6 Google Colab instances**, maximizing speed and fault tolerance, with output **pushed directly to Hugging Face**.

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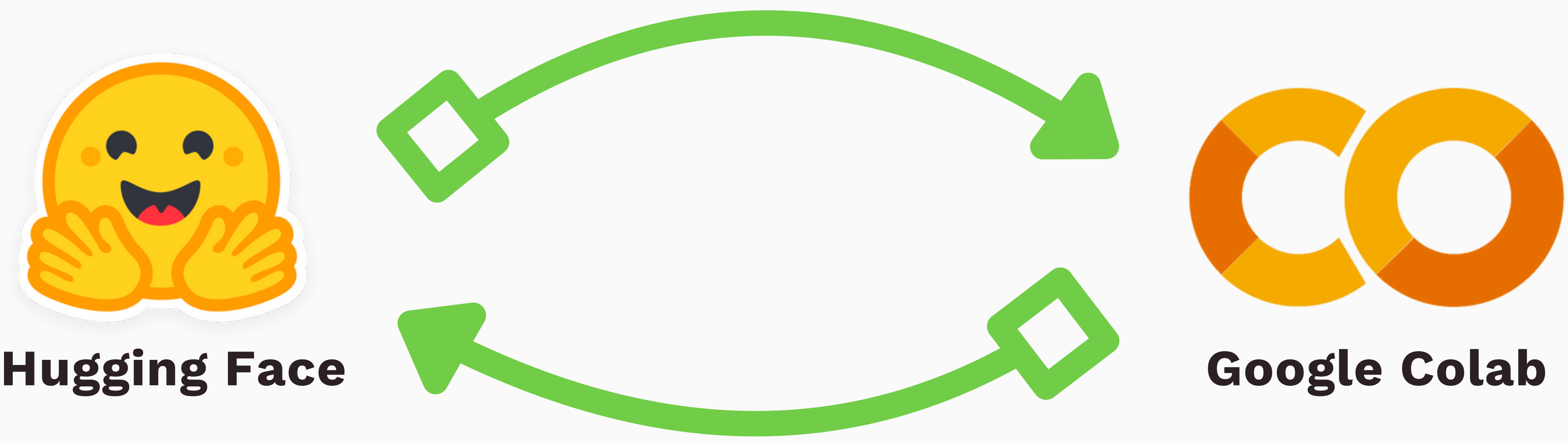
Dataset Metrics & Legal Content

Transparency in data collection is key for responsible R&D.

- **Total Documents**: ~250,000 successfully downloaded
- **Format**: Archived .zip files (each containing ~1,000 PDFs)
- **Time Cost**: ~200 hours of distributed scraping
- **Failure Rate**: Documented failures due to unreachable or invalid source URLs were excluded
- **Document Content**: The corpus covers the full hierarchy of Indonesian law, including UU, PP, Perpres, Permen, Pergub, and Perda, making it ideal for high-fidelity R&D
- **Repository**: [Azzindani/ID_REG]



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Efficiency in Legal Data Collection

- **Manual download by hire**: ~3,000 hrs (~18 months full-time) · ~6,000 USD labor cost
- **Manual download by myself** (4 hrs/day): 3,000 hrs (~29 months) ~150 USD electricity cost
- **Automated pipeline** (6 parallel processes on one PC): ~200 hrs (~8 days) ~10 USD electricity cost
- **Time saved**: ~93% vs full-time hire; >90% vs personal 4 hrs/day schedule
- **Cost saved**: ~99.8% vs hire; ~93% vs personal effort
- My workload reduced from months of **manual clicking → minimal setup & monitoring**
- **Future Plan**:
  1. **Apply OCR and PDF extraction** tools to improve text quality from scanned PDFs.
  2. Use this dataset as a ground base for building **regulation-aware RAG systems**, legal document classification, and retrieval tasks.
- **Acknowledgements**:
  1. Hugging Face
  2. Google Colab

This project demonstrates that large-scale, automated collection of Indonesian legal documents is feasible, efficient, and research-ready

Curious about building datasets efficiently with minimal effort?

Let’s talk—your ideas can shape the future of this open research.

Afza Azzindani